Benchmarking synthetic cold-start and temporal/provenance distribution shifts in drug–target interaction prediction

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Abstract

Reliable validation of drug-target interaction (DTI) models remains challenging because synthetic cold-start splits are often used as surrogates for prospective generalization. Here, we examine how benchmark design alters model ranking in DTI prediction. We evaluated a five-model comparison panel. It comprises a descriptor-based MLP reference model, a Random Forest trained on the same Morgan fingerprint/protein k-mer descriptor set, two pooled language-model baselines (`DTI-LM` and `HyperPCM`), and a graph-based `GraphDTA-style` baseline. Across three synthetic `BindingDB_Kd` splits, the highest-ranked model depended on the split definition: the Random Forest performed best on `unseen_drug` and `blind_start`, whereas the descriptor-based MLP remained strongest on `unseen_target`. We then evaluated temporal/provenance benchmarks derived from patent and non-patent BindingDB subsets and observed further ranking changes. A scaffold-restricted synthetic control sharpened the synthetic-side interpretation, while overlap-restricted patent variants showed that conclusions within the temporal family were themselves sensitive to overlap policy. Year-band and overlap-bucket analyses further indicated that temporal/provenance out-of-distribution behavior is structured rather than monolithic. To support reproducible evaluation, we release split manifests, benchmark tables, and a standalone evaluator. These results show that synthetic cold-start and temporal/provenance benchmarks are not interchangeable validation settings for DTI prediction, and that benchmark-qualified reporting is necessary when making model generalization claims.

Keywords: drug–target interaction, benchmark design, temporal validation, distribution shift, BindingDB

1 Introduction

Predicting drug-target interactions for unseen molecules or proteins is a central goal in computational drug discovery. Because truly prospective evaluation is difficult and expensive, many recent studies report held-out drug, target, or blind splits on datasets derived from BindingDB or DAVIS as proxies for inductive generalization [1, 2, 5–8, 13, 14]. These protocols are useful, but they also encourage a strong implicit assumption: if a method performs well on one cold-start split, then it is broadly robust to the kinds of shift encountered in deployment.

That assumption is weaker than it appears. Different split definitions remove different forms of overlap, and they may favor different model families. A pooled-LM baseline that looks strong under one synthetic exclusion rule may no longer be the most reliable option once the benchmark shifts in time, provenance, or assay composition. If these ranking changes occur even within a compact studied panel, benchmark design is not a technical footnote. It becomes part of the scientific claim.

This paper studies that question directly. Rather than proposing another DTI architecture, we examine whether model ranking is preserved across synthetic cold-start regimes and temporal/provenance shifts. We use a five-model comparison panel consisting of a descriptor-based MLP reference model (`base`), a Random Forest trained on the same Morgan fingerprint/protein k-mer descriptor set, two pooled language-model baselines (`DTI-LM` and `HyperPCM`), and a graph-based `GraphDTA-style` baseline. Even on the synthetic reference panel, rankings are already heterogeneous: `RF` is strongest on synthetic `unseen_drug` and `blind_start`, while `base` remains strongest on synthetic `unseen_target`.

We then contrast this synthetic picture with primary temporal/provenance benchmarks built from BindingDB patent and non-patent time-based splits. The resulting ordering changes again. The synthetic panel does not point to a single dominant system, and the temporal family introduces additional sensitivity to overlap policy within the same benchmark family. Paired bootstrap confidence intervals support the main temporal comparisons, but the broader claim is about benchmark-qualified ranking rather than any universal winner: benchmark definition materially changes the apparent ordering and therefore changes the conclusion a practitioner would draw about model generalization.

To move beyond leaderboard-style reporting, we also diagnose the primary temporal/provenance benchmark by year band and overlap bucket. This analysis shows that temporal/provenance OOD is not simply "harder cold-start." Instead, the gap is concentrated in specific year bands and overlap regimes, which helps explain why a method that looks strong under one split can lose under another. We further include matched 3-seed support on `BindingDB_Ki` and `DAVIS` to test whether the same model family remains stable across shifted target-cold settings. Recent molecular benchmark studies have made analogous arguments about OOD sensitivity in adjacent tasks [3, 4, 11]; our contribution is to bring that perspective into DTI as a reproducible validation study with reusable split manifests and a standalone evaluator.

Our main contributions are:

1. We construct a matched 3-seed synthetic reference panel on `BindingDB_Kd` and show that even standard cold-start evaluation is regime-dependent.
2. We establish primary temporal/provenance benchmarks on patent and non-patent BindingDB slices and show that their ranking can differ sharply from the synthetic panel.

3. We compare descriptor-based, pooled language-model, and graph-based baselines within one consistent five-model comparison panel, showing that benchmark sensitivity is not confined to one architecture family.
4. We provide year-band and overlap-bucket diagnoses, matched support on `BindingDB_Ki` and `DAVIS`, and reusable split manifests and evaluator resources to make the benchmark suite easy to inspect and reuse.

For practical model selection in DTI, these results imply that a single synthetic split is not sufficient evidence for generalization. At minimum, one synthetic axis and one temporal/provenance-shifted axis should be reported when models are compared for compound–target prioritization or related downstream use. The rest of the paper first defines the benchmark setup, then reports the synthetic and temporal/provenance comparisons, and finally discusses the implications for DTI validation practice.

2 Related Work

Synthetic cold-start evaluation in DTI. Machine-learning models for DTI and affinity prediction are commonly evaluated on BindingDB- and DAVIS-derived random, cold-drug, cold-target, or blind splits [1, 2, 5–8, 13, 14]. These protocols are useful for controlled comparison, but they eliminate different forms of chemical, protein, and pair overlap and therefore do not represent a single validation setting. For this reason, results obtained on one synthetic split should not automatically be interpreted as evidence of general robustness.

Validation design in chemoinformatics. More broadly in chemoinformatics, validation design and test-set construction are known to influence apparent model performance. Reanalysis of large bioactivity benchmarks, increasingly challenging potency-prediction partitions, and transfer-learning studies for binary drug–target interactions have all shown that the relative standing of deep and classical models can change materially when the evaluation setting changes [9, 10, 12]. Our study follows

this validation-oriented tradition by asking how split semantics alter conclusions in a DTI setting built around widely used BindingDB and DAVIS resources.

Benchmark and resource studies under distribution shift. Recent benchmark studies in molecular machine learning and related biomedical prediction problems have shown that scaffold, temporal, and domain-shifted evaluation can change model ranking and alter the practical interpretation of benchmark results [3, 4, 11]. This literature has shifted attention from purely i.i.d. comparison toward reusable benchmark resources that expose how conclusions depend on data partitioning. Our work extends that perspective to DTI by coupling multiple benchmark families with exported split manifests and a standalone evaluator.

Positioning. Prior DTI papers typically optimize performance within one benchmark design. In contrast, our contribution is a reproducible case study of ranking stability across synthetic cold-start and temporal/provenance benchmarks, together with overlap-sensitive robustness analyses and reusable evaluation resources. The emphasis is therefore on validation methodology and benchmark semantics rather than on proposing a new predictive architecture.

3 Benchmark Setup

Our goal is to test whether model ranking is preserved across synthetic cold-start regimes and real distribution shifts. We therefore use a five-model comparison panel. `base` is a descriptor-based MLP reference system built from RDKit Morgan fingerprints and hashed protein k-mer features. `RF` is a Random Forest baseline trained on the same descriptor set as `base`, which allows the effect of the learning algorithm to be separated from the effect of the representation. `DTI-LM` is a pooled ChemBERTa+ESM2 baseline aligned to recent literature, `HyperPCM` is a task-conditioned pooled-LM baseline, and `GraphDTA-style` is a graph-based baseline with a molecular GCN and protein CNN. Together these models span descriptor-based neural and classical learners, pooled

Table 1 Benchmark taxonomy and seed0 split statistics. Test prevalence differs substantially across benchmark families, so AUPRC should be compared within benchmark rather than across rows.

Benchmark	Shift family	Synth.?	Test rows	Pos. rate
`Kd / unseen_drug`	drug-cold	yes	12,744	0.1966
`Kd / blind_start`	drug+target-cold	yes	2,821	0.2031
`Kd / unseen_target`	target-cold	yes	12,052	0.2239
Patent temporal/prov.	temporal/prov.	no	49,028	0.5745
Nonpatent temporal/prov.	temporal/prov.	no	7,174	0.4260
`Ki / unseen_target`	assay+target-cold	mixed	70,380	0.4652
`DAVIS / unseen_target`	dataset+target-cold	mixed	5,168	0.0615

language-model baselines, and a graph-based baseline. Because the anchor system is central to every ranking claim, Appendix Table C4 gives the exact `base` feature construction, architecture, parameter count, and training recipe.

We evaluate this comparison panel on two benchmark groups. The first is a synthetic reference panel based on `BindingDB_Kd`, with three split definitions: `unseen_drug`, `blind_start`, and `unseen_target`. The second comprises temporal/provenance and external-shift benchmarks. Our primary temporal/provenance analyses are two year-based BindingDB benchmarks: a patent split (`patent_temporal`) and a non-patent split (`nonpatent_temporal`). Both are evaluated over three seeds. We also include `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting shifted target-cold benchmarks that test whether the same model family remains stable across assay and dataset shifts. Table 1 summarizes the shift taxonomy and split statistics.

We report AUPRC as the primary metric and AUROC as a secondary metric. AUPRC is emphasized because within each benchmark the central question is imbalanced ranking quality, and precision–recall behavior is the quantity most directly tied to the benchmark-sensitive model-selection claims made here. Because prevalence ranges from 0.0615 on `DAVIS` to 0.5745 on `patent_temporal`, we do not interpret absolute AUPRC values across benchmark families. Instead, AUPRC is compared within benchmark, while AUROC is retained as a prevalence-insensitive secondary check. For very low-prevalence settings such as `DAVIS`, enrichment-oriented metrics such as enrichment factor or BEDROC would add useful screening-specific context; we leave these

to future benchmark extensions so that all panels in this study can still be compared under one consistent metric pair. We report matched three-seed results on the synthetic reference panel and on the primary temporal/provenance benchmarks. The supporting `BindingDB_Ki` and `DAVIS` panels are also reported with matched three-seed evaluation, but they are interpreted as supporting evidence rather than as co-equal headline axes. Paired bootstrap confidence intervals are used for the principal ranking comparisons. Appendix Table G8 collects these intervals.

To make the benchmark reusable, each split is exported as a canonical resource bundle with manifests, split tables, cached identifiers, and a standalone CSV-based evaluator. This resource layer allows external predictions to be scored against the exact same test partitions used in the paper. In practical terms, an external user only needs a split bundle plus a prediction CSV keyed by `example_id`; the reported metrics can then be reproduced with `python scripts/evaluate_prediction_csv.py --manifest manifest.json --predictions preds.csv`. Appendix Table H9 enumerates the released artifacts and how they are consumed by the evaluator.

The patent benchmark requires several setup choices that are documented explicitly. We use a local year-based split of `BindingDB_patent`, with training years before the validation year, a held-out validation year, and a later temporal test set. Because this benchmark also changes provenance and assay composition, we describe it as temporal/provenance OOD rather than pure time shift. We also use a patent-specific affinity conversion rather than reusing the standard `pKd` transform, because the patent benchmark stores affinities in a different log-scale convention. Appendix Table A1 summarizes the exact year boundaries, split counts, label construction, overlap policy, and seed semantics for both `patent_temporal` and the alternative-cutoff `patent_temporal_v2017`. Appendix Table B3 provides the same information for the additional `BindingDB_nonpatent_Kd / nonpatent_temporal` temporal analysis.

BindingDB-derived panels are anchored in the BindingDB resource [6], while DAVIS provides an external kinase-focused support benchmark [2].

Finally, for diagnostic analysis, we use the saved overlap score from each test prediction to compute overlap buckets that summarize how strongly each test example resembles the training support. Concretely, `overlap_score` is defined as $\min(s_d, s_t)$, where s_d is the cosine similarity between the test drug's Morgan fingerprint and its nearest training-drug neighbor, and s_t is the cosine similarity between the test target's hashed k-mer feature vector and its nearest training-target neighbor. We use the minimum to conservatively define overlap: a pair is high-overlap only when both its drug and target are well represented in the training set. We retain cosine for the released overlap score so that drug-side and target-side overlap share the same similarity scale inside this joint diagnostic; the separate synthetic similarity analysis still reports the standard Morgan-fingerprint Tanimoto and scaffold-overlap statistics. We report benchmark taxonomy and split statistics, including test-set size and prevalence, so that AUPRC is interpreted within benchmark rather than across benchmark families.

4 Results

4.1 Synthetic cold splits are already regime-dependent

We begin with the `BindingDB_Kd` synthetic reference panel. Table 2 shows the main pattern: even within a single dataset, model rankings are not stable across split definitions. On `unseen_drug`, the highest mean AUPRC is obtained by `RF`, which reaches 0.7339 ± 0.0051 versus 0.5509 ± 0.0900 for `base`, 0.5694 ± 0.0473 for `HyperPCM`, 0.5004 ± 0.0341 for `DTI-LM`, and 0.5626 ± 0.0172 for `GraphDTA-style`. The paired bootstrap interval for `RF - base` is strictly positive, and `RF` also remains above `HyperPCM` under paired bootstrap. On `blind_start`, `RF` again reaches the highest mean AUPRC at 0.4512 ± 0.0373, compared with 0.3910 ± 0.0504 for `GraphDTA-style`, 0.3828 ± 0.1024 for `HyperPCM`, and 0.3737 ± 0.0229 for `base`, and the paired bootstrap interval for `RF -`

base remains strictly positive. On unseen_target, however, the synthetic target-cold story is more stable. base remains strongest by mean AUPRC at 0.5224 ± 0.0602, while RF is close but slightly lower at 0.5183 ± 0.0612, and the paired bootstrap interval for RF - base crosses zero.

These results show that "synthetic cold-start DTI" is already not a single validation condition. Different synthetic regimes reward different model families, and the five-model comparison panel shows that this is not merely a pooled-language-model phenomenon. A Random Forest using the same descriptor set as base becomes strongest on two synthetic axes, while the graph-based baseline remains competitive but does not dominate. Additional similarity and scaffold analyses sharpen this point further: on synthetic unseen_drug, the held-out compounds still have median nearest-train Tanimoto 0.7732 and Murcko scaffold match rate 0.6204. When we replace entity holdout with a stricter scaffold split, the earlier HyperPCM edge is no longer supported. Over three seeds on BindingDB_Kd / scaffold_drug, base reaches $0.6069 \pm 0.0161/0.8351 \pm 0.0037$, while HyperPCM reaches $0.5983 \pm 0.0368/0.8155 \pm 0.0204$, and the paired bootstrap interval for HyperPCM - base crosses zero (Appendix Tables F7 and G8). The synthetic panel therefore remains useful, but both its drug-cold ordering and its apparent best learner are sensitive to how chemically strict the split actually is.

4.2 Primary temporal/provenance benchmarks alter the ranking again

We next evaluate the same five-model comparison panel on the primary temporal/provenance benchmarks. On BindingDB_patent / patent_temporal, RF attains the highest mean AUPRC, reaching 0.8234 ± 0.0012, compared with 0.7772 ± 0.0086 for base, 0.7825 ± 0.0078 for DTI-LM, 0.7753 ± 0.0050 for HyperPCM, and 0.7726 ± 0.0016 for GraphDTA-style. The paired bootstrap intervals for RF - base and RF - DTI-LM remain strictly positive. The key observation is therefore not merely that RF wins

Table 2 Synthetic reference panel and primary temporal/provenance benchmarks for the five-model comparison panel. Benchmarks are shown as columns and models as rows to improve legibility in the journal layout. Each cell reports mean ± std over seeds, with AUPRC on the first line and AUROC on the second line. Best mean AUPRC in each benchmark column is bold. `RF` is a Random Forest baseline trained on the same Morgan+k-mer descriptor set as `base`, and `GraphDTA-style` is the graph-based baseline. All entries are matched three-seed summaries.

Model	BindingDB$_{Kd}$ drug-cold	BindingDB$_{Kd}$ blind-start	BindingDB$_{Kd}$ target-cold	Patent temporal/prov.	Nonpatent temporal/prov.
	0.5509 ± 0.0900	0.3737 ± 0.0229	**0.5224 ± 0.0602**	0.7772 ± 0.0086	0.6369 ± 0.0173
`base`	0.7794 ± 0.0534	0.6778 ± 0.0151	**0.7741 ± 0.0161**	0.7223 ± 0.0109	0.7333 ± 0.0081
	0.7339 ± 0.0051	**0.4512 ± 0.0373**	0.5183 ± 0.0612	**0.8234 ± 0.0012**	**0.7201 ± 0.0012**
`RF`	**0.8854 ± 0.0088**	**0.7451 ± 0.0260**	0.7838 ± 0.0218	**0.7773 ± 0.0017**	**0.7627 ± 0.0020**
	0.5004 ± 0.0341	0.3636 ± 0.0693	0.5136 ± 0.0730	0.7825 ± 0.0078	0.6531 ± 0.0066
`DTI-LM`	0.7364 ± 0.0122	0.6490 ± 0.0642	0.7970 ± 0.0159	0.7362 ± 0.0083	0.7644 ± 0.0041
	0.5694 ± 0.0473	0.3828 ± 0.1024	0.4955 ± 0.0793	0.7753 ± 0.0050	0.6287 ± 0.0041
`HyperPCM`	0.7778 ± 0.0448	0.6807 ± 0.0824	0.7810 ± 0.0191	0.7368 ± 0.0090	0.7450 ± 0.0088
	0.5626 ± 0.0172	0.3910 ± 0.0504	0.4722 ± 0.0925	0.7726 ± 0.0016	0.6635 ± 0.0142
`GraphDTA-style`	0.7988 ± 0.0129	0.7141 ± 0.0309	0.7637 ± 0.0314	0.7232 ± 0.0007	0.7334 ± 0.0096

this benchmark, but that the ranking differs from the synthetic reference panel and therefore depends on benchmark semantics.

A second temporal analysis strengthens this reading further. On the non-patent benchmark built directly from the official BindingDB release, `RF` again attains the highest mean AUPRC over matched 3-seed evaluation, reaching $0.7201 \pm 0.0012/0.7627 \pm 0.0020$, versus $0.6635 \pm 0.0142/0.7334 \pm 0.0096$ for `GraphDTA-style`, $0.6531 \pm 0.0066/0.7644 \pm 0.0041$ for `DTI-LM`, $0.6369 \pm 0.0173/0.7333 \pm 0.0081$ for `base`, and $0.6287 \pm 0.0041/0.7450 \pm 0.0088$ for `HyperPCM`. Here too, the paired bootstrap intervals stay strictly positive for both `RF - base` and `RF - DTI-LM`. Together, the patent and non-patent analyses show that temporal/provenance evaluation does not preserve the ordering seen on the synthetic panel.

At the same time, two controlled patent variants show that this temporal ordering is not immutable once overlap is constrained more aggressively. In the pair-novel variant (`patent_temporal_pair_novel`), `DTI-LM` becomes the strongest AUPRC model at 0.7491 ± 0.0076, ahead of `RF` at 0.7393 ± 0.0015 and `base` at 0.7378 ± 0.0107. The paired bootstrap interval for `DTI-LM - RF` is strictly positive, while `RF - base` crosses zero. In the drug-novel variant (`patent_temporal_drug_novel`), `DTI-LM` again remains strongest by AUPRC at 0.7402 ± 0.0077, versus 0.7356 ± 0.0016 for `RF`, with a paired interval for `DTI-LM - RF` that also stays positive. These additional robustness analyses make the temporal conclusion more precise rather than weaker: the patent benchmark is sensitive not only to benchmark family but also to how much entity or pair recurrence is allowed inside a nominally temporal split. Figure 1 is therefore a compact diagnostic reduction of Table 2 rather than a second copy of the full five-model panel: it shows only deltas against `base`, so `base` itself defines the zero line, and it restricts the display to `RF`, `DTI-LM`, and `HyperPCM` as compact representatives of the descriptor-based, pooled-language-model, and hybrid comparison families.

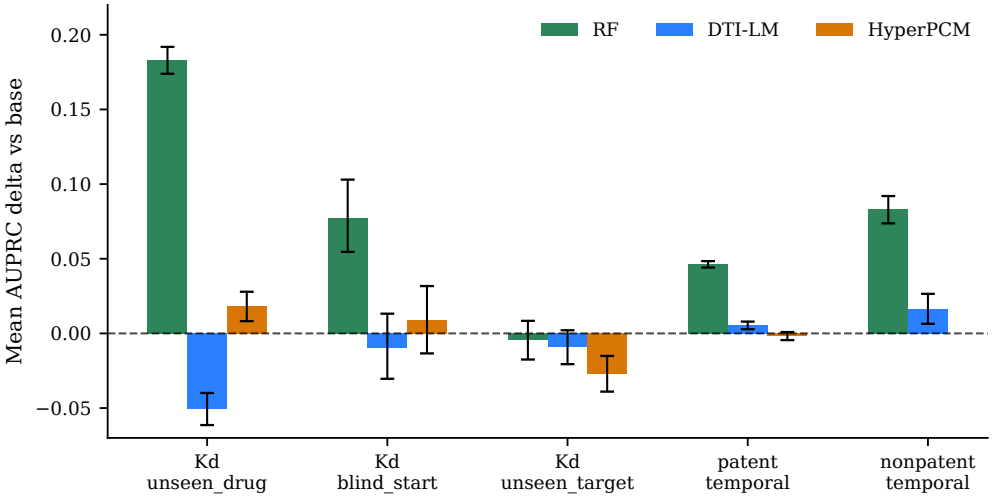

Fig. 1 Compact within-benchmark AUPRC delta summary relative to `base` for the five-model comparison panel. Because the figure is a delta-to-`base` diagnostic, `base` defines the zero line and is not drawn as a bar series. The corresponding `base` AUPRC values are 0.5509 for `BindingDB_Kd / unseen_drug`, 0.3737 for `blind_start`, 0.5224 for `unseen_target`, 0.7772 for `patent_temporal`, and 0.6369 for `nonpatent_temporal`. We show `RF`, `DTI-LM`, and `HyperPCM` as compact representatives of the descriptor-based, pooled-language-model, and hybrid comparison families; `GraphDTA-style` remains reported in Table 2 but is omitted here to keep the delta panel readable. Because prevalence differs across benchmarks, the vertical scale should be read as a row-wise within-benchmark contrast rather than as a cross-benchmark effect-size scale

This is the central result. The synthetic panel does not produce a single best model, and the primary temporal/provenance benchmarks do not preserve the same ordering either. Instead, both benchmark axis and model family change the apparent ranking. A narrower earlier-cutoff probe also points in the same direction: under `patent_temporal_v2017`, the matched 3-seed two-model ordering `DTI-LM > base` is preserved (Appendix Table A2), which suggests that the patent-family sensitivity is not driven only by the 2018/2019 boundary. The five-model comparison panel and the controlled patent variants nevertheless carry the main argumentative weight, because they show more directly that the apparent best system can change not only across benchmarks but also across overlap policies inside the same temporal family.

Table 3 Compact AUPRC summary behind the patent diagnosis for the five-model comparison panel. For readability, this diagnostic table focuses on `base`, `RF`, and `DTI-LM`, the three systems that determine the aggregate and slice-level ranking changes discussed in the text; `HyperPCM` and `GraphDTA-style` remain listed in Table 2. Values are mean ± seed std over the matched 3-seed runs. The tiny 2021 slice is pooled into the `2020--2021` stratum to avoid over-interpreting a 72-example cell.

Slice type	Slice	Count	base	RF	DTI-LM
Year	2019	42,009	0.7626 ± 0.0098	**0.8062 ± 0.0013**	0.7734 ± 0.0071
Year	2020–2021	7,019	0.8716 ± 0.0121	**0.9098 ± 0.0005**	0.8533 ± 0.0100
Overlap bucket	0 (0.000–0.531)	16,341	0.5551 ± 0.0190	0.6005 ± 0.0034	**0.6353 ± 0.0175**
Overlap bucket	1 (0.531–0.710)	16,342	0.7218 ± 0.0167	**0.7606 ± 0.0012**	0.7489 ± 0.0094
Overlap bucket	2 (0.710–0.989)	16,345	0.9237 ± 0.0030	**0.9426 ± 0.0005**	0.8866 ± 0.0097

4.3 Patent temporal/provenance shift is structured across year bands and overlap levels

To help explain why the temporal/provenance benchmark changes the ranking, we analyze the patent results by year band and by overlap bucket. Table 3 and Figure 2 are intentionally compact: they focus on `base`, `RF`, and `DTI-LM`, because these three systems define the patent ranking story in the aggregate benchmark, the lowest-overlap bucket, and the controlled overlap variants. `HyperPCM` and `GraphDTA-style` remain reported in Table 2, but neither leads a year-band or overlap-bucket slice. To avoid over-reading the tiny 2021 cell, we pool 2020 and 2021 into a single recent stratum. Within the five-model comparison panel, `RF` is strongest in both the large 2019 slice and the pooled 2020–2021 slice, so the year-band view does not support a single uniform explanation based on one model family alone.

The overlap-bucket analysis reveals another useful pattern. In the lowest-overlap bucket, `DTI-LM` remains the strongest system, but `RF` becomes strongest in the middle- and high-overlap buckets and therefore dominates the aggregate temporal benchmark. The structured-shift diagnosis is therefore reinforced: temporal/provenance OOD is not just a colder version of synthetic cold-start, because different overlap strata continue to favor different model families. The benchmark outcome depends on both the shift axis and the modeling family, not only on whether a model uses pooled language-model features.

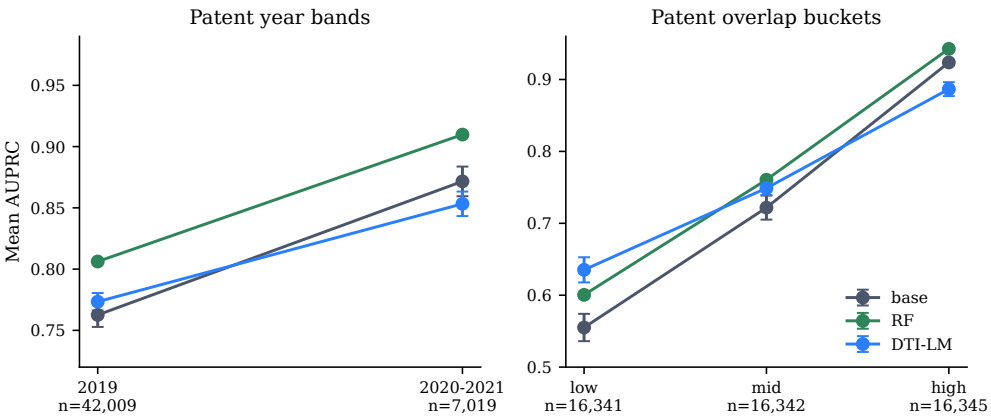

Fig. 2 Patent temporal/provenance diagnosis in terms of mean AUPRC for the five-model comparison panel. For readability, the plot focuses on `base`, `RF`, and `DTI-LM`, which are the three systems that define the aggregate and slice-level ranking changes in the patent analyses; `HyperPCM` and `GraphDTA-style` remain listed in Table 2 but do not lead a year-band or overlap-bucket slice. Left: pooled year-band slices, with 2021 merged into the recent stratum. Right: test-set overlap buckets computed from the saved overlap score. Error bars denote seed std over the matched 3-seed runs. `RF` dominates the aggregate temporal benchmark, while `DTI-LM` remains strongest only in the lowest-overlap bucket

4.4 Supporting target-cold OOD benchmarks remain heterogeneous

We treat `BindingDB_Ki / unseen_target` and `DAVIS / unseen_target` as supporting real-OOD benchmarks rather than primary headline axes. Under matched three-seed evaluation, they extend the benchmark-sensitivity story rather than collapse it to a single winner. On `BindingDB_Ki`, `base` remains strongest at $0.6934 \pm 0.0257/0.7192 \pm 0.0068$, ahead of `DTI-LM` at $0.6648 \pm 0.0177/0.6936 \pm 0.0044$, `HyperPCM` at $0.6515 \pm 0.0301/0.6757 \pm 0.0211$, `RF` at $0.6328 \pm 0.0465/0.6581 \pm 0.0319$, and `GraphDTA-style` at $0.6280 \pm 0.0237/0.6550 \pm 0.0193$. On `DAVIS`, however, the ordering changes again: `RF` attains the highest mean AUPRC at $0.4408 \pm 0.0325/0.8806 \pm 0.0128$, ahead of `base` at $0.4097 \pm 0.0275/0.8613 \pm 0.0153$, `DTI-LM` at $0.4090 \pm 0.0570/0.8718 \pm 0.0152$, `HyperPCM` at $0.3840 \pm 0.0269/0.8610 \pm 0.0245$, and `GraphDTA-style` at $0.3601 \pm 0.0502/0.8359 \pm 0.0292$.

These auxiliary results are therefore consistent with the main claim of the paper, but in a more informative way than the earlier pooled-LM-only view. `BindingDB_Ki` and

DAVIS do not recover a stable winner. Instead, they show that even among supporting target-cold OOD benchmarks, the apparent best family can differ by benchmark definition. They still carry less argumentative weight than the patent benchmark because they do not provide a cleaner shift axis than temporal/provenance OOD, but they reinforce the same benchmark-qualified interpretation of model generalization.

5 Discussion

The main message of this study is that DTI evaluation is benchmark-sensitive in a way that current reporting practice often obscures. A synthetic split can still be useful, but it should not be treated as a universal proxy for real OOD generalization. Within the five-model comparison panel studied here, the sign and magnitude of an apparent gain change when the benchmark definition changes, and temporal/provenance conclusions can change again when overlap policy is tightened within the same benchmark family.

For chemoinformatics practice, the implication is straightforward. A single synthetic cold split is not sufficient support for a generalization claim, and model selection can depend on whether evaluation emphasizes entity holdout, time/provenance shift, or overlap restriction. Three practical reporting recommendations follow from the present results. First, DTI studies should report at least one synthetic split together with one temporal/provenance-shifted benchmark when model generalization is discussed. Second, studies should disclose overlap characteristics, because entity or pair recurrence can materially change the apparent ranking even within one temporal benchmark family. Third, claims about the superiority of descriptor-based, pooled language-model, or graph-based models should be phrased in benchmark-qualified terms rather than as architecture-level general statements.

These recommendations are directly relevant to routine chemoinformatics workflows. When DTI models are used for compound–target prioritization, virtual screening support, or internal model selection, the preferred model can differ depending on

whether validation resembles entity holdout or time/provenance shift. In that setting, benchmark diversity is not an aesthetic addition to the manuscript; it is part of the validation protocol needed to avoid overconfident selection decisions.

One empirical pattern also deserves a mechanism-level interpretation. `RF` is repeatedly strong on synthetic `unseen_drug`, `blind_start`, and both temporal/provenance panels, yet weaker on synthetic and external target-cold settings. Because `RF` and `base` use the same Morgan+k-mer descriptor set, this pattern is not about representation alone. It suggests that when held-out drugs remain close to seen scaffolds, or when temporal/provenance shift still preserves substantial drug-side support, tree ensembles can exploit local descriptor neighborhoods effectively without the additional smoothing assumptions imposed by the MLP. When the dominant challenge shifts toward protein-side extrapolation, as in `unseen_target` and `BindingDB_Ki`, the hashed k-mer representation appears to become the bottleneck and the earlier `RF` advantage disappears. We therefore interpret the recurrent `RF` gains as evidence of strong descriptor-space interpolation under drug-side support, not as a universal advantage of classical learners over neural models.

The study also has clear limitations. Although the five-model comparison panel spans descriptor-based neural and classical learners, pooled language-model baselines, and a graph-based baseline, it is still not exhaustive. Important recent architectures such as ColdstartCPI, DrugLAMP, and SP-DTI are not part of the present panel [5, 7, 14]. Our conclusion should therefore be read as a methodological one rather than as a universal leaderboard claim: ranking instability is already visible in a compact but deliberately diverse panel, but a larger community benchmark would be needed to quantify how far this effect extends across the full modern DTI literature.

The patent benchmark also mixes temporal, provenance, and assay shifts, so it should be read as a structured temporal/provenance stress test rather than as a clean causal estimate of time shift alone. The year-band and overlap-bucket analyses show

that behavior within this compound shift is heterogeneous, but they do not identify whether chronology, source curation, or assay composition is the dominant driver. The non-patent temporal benchmark partially reduces the provenance confound and preserves a ranking change, which argues against a patent-only artifact, but its test set is much smaller (7,174 versus 49,028 rows) and therefore cannot provide a one-to-one decomposition. A cleaner attribution would require larger source-matched temporal cohorts or prospective snapshots with harmonized assay composition.

Finally, the metric and uncertainty design should be read with their intended scope. AUPRC is appropriate for within-benchmark model ranking under class imbalance, but it is not meant to support direct comparisons of absolute difficulty across benchmark families; for very low-prevalence settings such as `DAVIS`, enrichment-style metrics would be a useful screening-oriented extension. Likewise, three matched seeds are sufficient for the present validation study but not generous. We used them to balance compute against coverage of multiple benchmark families, and the paired bootstrap intervals provide example-level uncertainty for the principal ranking claims. Even so, high-variance rows such as synthetic `unseen_drug` indicate that larger seed counts or an explicit power analysis would strengthen future versions of the benchmark. We also do not evaluate prospective biological decision quality. These limitations do not weaken the central conclusion. They instead define its scope: benchmark definition changes model ranking in DTI, and even temporal/provenance conclusions depend on overlap policy.

6 Conclusion

We presented a validation-oriented study of distribution shift in DTI prediction. Across a five-model comparison panel, a 3-seed synthetic reference panel on `BindingDB_Kd` showed that cold-start regimes are already heterogeneous, while temporal/provenance benchmarks built from patent and non-patent BindingDB subsets changed the ranking

again. Additional scaffold-restricted analysis sharpened the synthetic-side interpretation, and overlap-restricted patent variants further showed that conclusions within the temporal family depend on overlap policy. Year-band and overlap-bucket analyses indicated that temporal/provenance OOD behavior is structured rather than monolithic. Overall, these results show that synthetic cold-start and temporal/provenance benchmarks are not interchangeable validation settings for DTI prediction. Future DTI studies should therefore report both synthetic and temporal/provenance evidence when making model-generalization claims and when selecting models for practical chemoinformatics workflows.

Declarations

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Competing interests The authors declare that they have no known financial or non-financial competing interests relevant to this study.

Author contributions Author names and contribution details will be finalized at submission.

Data availability A fixed review snapshot corresponding to git commit `a5d6b8b` (`a5d6b8bd96c98fed73c5a9771c2a1079c9f12255`) is archived with the submission and mirrored in the project repository at https://github.com/cicisoso/Synthetic-Cold-Splits-Are-Not-Enough. The snapshot includes the exported benchmark resources under `benchmark_resources/`, the benchmark tables under `reports/`, and the release manifest at `benchmark_resources/RELEASE_MANIFEST.md`. A DOI-backed public archive will be deposited at acceptance.

Code availability The exact code used to construct the benchmark splits, train the comparison models, and reproduce the reported analyses corresponds to commit `a5d6b8b`. The evaluator entry points are `scripts/evaluate_prediction_csv.py` and

`scripts/evaluate_decision_csv.py`; the full release contents are enumerated in `benchmark_resources/RELEASE_MANIFEST.md`.

Table A1 Specification of the primary patent temporal/provenance benchmarks. The year partition is fixed; the random seed changes training stochasticity only.

Aspect	`patent_temporal`	`patent_temporal_v2017`
Source files	`train_val.csv` + `test.csv`	`train_val.csv` + `test.csv`
Train years	2013–2017	2013–2016
Validation year	2018	2017
Test years	2019–2021	2018–2021
Rows (train / valid / test)	142,275 / 41,155 / 49,028	90,771 / 51,504 / 90,183
Pos. rate (train / valid / test)	0.566 / 0.534 / 0.575	0.568 / 0.562 / 0.556
Label transform	patent log-scale transform $Y \mapsto \mathrm{p}K_d = 9 - Y/\ln(10)$; positive if $\mathrm{p}K_d \geq 7.0$	
Overlap policy	entity overlap allowed; similarity exported through `overlap_score`	
Duplicate policy	pair rows retained after year split; entity tables deduplicated by canonical ID	
Seed semantics	year partition fixed; seed changes training stochasticity only	
OOD interpretation	temporal/provenance OOD, not pure time shift	

Table A2 Earlier-cutoff patent robustness probe. This auxiliary table reports the matched 3-seed `base` versus `DTI-LM` comparison on `patent_temporal_v2017`. It is included to document the evidence behind the earlier-cutoff statement in the main text, not to replace the five-model main panel.

Benchmark	`base` AUPRC AUROC	`DTI-LM` AUPRC AUROC	Reading
`BindingDB_patent`	0.7044 ± 0.0058	0.7134 ± 0.0027	
`patent_temporal_v2017`	0.6624 ± 0.0079	0.6754 ± 0.0039	earlier cutoff preserves the same `DTI-LM > base` ordering as the main patent-family robustness screen

Appendix A Patent benchmark specification

Because the main paper depends on `BindingDB_patent / patent_temporal`, we make its construction explicit here. Table A1 enumerates the year boundaries, split counts, label construction, overlap policy, and seed semantics for both the main patent split and the earlier-cutoff robustness variant.

Appendix B Non-patent temporal probe specification

Because the main revision also relies on `BindingDB_nonpatent_Kd / nonpatent_temporal` to separate temporal drift from patent provenance, we make that construction explicit as well. Table B3 lists the source file, year cutoffs, label

Table B3 Specification of the non-patent temporal probe. The year partition is fixed; the random seed changes training stochasticity only.

Aspect	`nonpatent_temporal`
Source file	official BindingDB TSV, filtered to rows without a patent number
Date field	publication year from `Date of publication`
Train years	1977–2018
Validation year	2019
Test years	2020–2023
Rows (train / valid / test)	35,755 / 3,006 / 7,174
Pos. rate (train / valid / test)	0.341 / 0.445 / 0.426
Label transform	raw K_d in nM with $\mathrm{p}K_d = -\log_{10}(Y/10^9)$; positive if $\mathrm{p}K_d \geq 7.0$
Overlap policy	entity overlap allowed; `drug_overlap`, `target_overlap`, and `overlap_score` exported
Duplicate policy	exact duplicate rows removed before partitioning
Seed semantics	year partition fixed; seed changes training stochasticity only
OOD interpretation	non-patent temporal shift with less provenance mixing than `patent_temporal`

transform, prevalence, duplicate policy, and seed semantics for the released non-patent temporal split.

Appendix C Reference-model specification

Because every ranking claim is anchored against `base`, we make that system explicit rather than treating it as an unnamed baseline. Table C4 specifies the exact input features, network structure, parameter count, optimization recipe, and implementation path used in every experiment.

Appendix D Random Forest and graph-based baseline specifications

Because the five-model comparison panel depends on the `RF` baseline and the `GraphDTA-style` graph-based baseline, we make those systems explicit as well. Table D5 mirrors the level of detail used for `base`, and also records the core `RF` hyperparameters used in the reported experiments.

Table C4 Specification of the `base` reference model used throughout the benchmark. This table is included to make the anchor system reproducible rather than treating it as an informal baseline label.

Component	Specification
Drug input	RDKit Morgan fingerprint, radius 2, 2048 bits, generated from canonical SMILES in `src/raicd/featurizers.py`.
Target input	L2-normalized hashed amino-acid 1/2/3-mer count vector, 2048 dimensions, generated in `src/raicd/featurizers.py`.
Drug encoder	Two-layer MLP: Linear$(2048, 512)$, GELU, dropout 0.1, Linear$(512, 256)$, followed by L2 normalization.
Target encoder	Same architecture as the drug encoder, with independent parameters.
Pair head	Concatenate $[d,\ t,\ d \odot t,\ \lvert d - t \rvert] \in \mathbb{R}^{1024}$, then Linear$(1024, 512)$, GELU, dropout 0.1, Linear$(512, 1)$.
Trainable parameters	2,886,145 parameters in total.
Optimization	BCE-with-logits objective, Adam with learning rate 10^{-3} and weight decay 10^{-4}, 8 epochs, batch size 256.
Labeling	Binary label derived from a `pKd` threshold of 7.0 (or patent-specific log-nM conversion for `BindingDB_patent`; see Table A1).
Code path	Model implementation in `src/raicd/model.py`; training entry point in `scripts/train_raicd.py`.

Appendix E Additional real-OOD support benchmarks

The main paper focuses on the strongest evidence chain: the synthetic `BindingDB_Kd` reference panel, the primary temporal/provenance benchmarks, and the patent year/overlap diagnosis. `BindingDB_Ki` and `DAVIS` remain supporting real-OOD benchmarks, but they now also use the full matched five-model comparison panel. This broader support table is helpful because it shows that the auxiliary target-cold OOD benchmarks remain heterogeneous as well: `base` stays strongest on `BindingDB_Ki`, whereas `RF` becomes strongest on `DAVIS`.

On the `BindingDB_Ki` support split, Table E6 shows that `base` remains ahead of the other four systems. On `DAVIS`, by contrast, `RF` becomes the strongest support-panel model by both AUPRC and AUROC, while `base` and `DTI-LM` remain close to one another. Together, these support panels reinforce the same message as the primary analyses: strong baselines do not transfer uniformly across OOD definitions, and even the auxiliary target-cold benchmarks do not support a single stable model ranking.

Table D5 Specification of the Random Forest and graph-based baselines. This table mirrors Table C4 so the non-reference baselines can be reproduced from the same level of detail.

Component	Specification
GraphDTA-style drug input	RDKit molecular graph from canonical SMILES. Node features encode atom symbol, degree, hydrogen count, valence, formal charge, aromaticity, and hybridization in `src/raicd/graph_baselines.py`.
GraphDTA-style drug encoder	Three-layer GCN: GCNConv$(36, 128)$, GCNConv$(128, 128)$, GCNConv$(128, 128)$, each followed by ReLU and dropout 0.1. Global mean and global max pooling are concatenated to produce a 256-dimensional drug embedding.
GraphDTA-style target input	Amino-acid token sequence with vocabulary size 21 (20 residues plus padding), truncated or zero-padded to length 1000.
GraphDTA-style target encoder	Embedding dimension 64, followed by three 1D convolutions Conv1d$(64, 128, k = 8)$, Conv1d$(128, 128, k = 8)$, and Conv1d$(128, 128, k = 8)$. Each block uses ReLU and dropout 0.1; global max pooling yields a 128-dimensional target embedding.
GraphDTA-style pair head	Concatenate the 256-dimensional drug embedding and 128-dimensional target embedding, then apply Linear$(384, 256)$, ReLU, dropout 0.1, Linear$(256, 128)$, ReLU, dropout 0.1, Linear$(128, 1)$.
GraphDTA-style trainable parameters	499,009 parameters in total under the default configuration used in the paper.
GraphDTA-style optimization	BCE-with-logits objective with train-split positive-class weighting, AdamW with learning rate 10^{-3} and weight decay 10^{-4}, 8 epochs, batch size 128.
RF core hyperparameters	`RandomForestClassifier` with `n_estimators`= 400, `max_depth`=None, `min_samples_leaf`= 1, `max_features`=`sqrt`, `class_weight`=`balanced_subsample`, and seed-specific `random_state`.
Code path	Graph baseline in `src/raicd/graph_baselines.py` with training in `scripts/train_graph_baseline.py`; Random Forest baseline in `src/raicd/classical_baselines.py` with training in `scripts/train_classical_baseline.py`.

Appendix F Additional robustness analyses

Table F7 records the additional analyses used to test whether the main conclusion depends on provenance or overlap policy. The first analysis constructs a non-patent temporal benchmark directly from the official BindingDB release. The second replaces entity holdout with a Murcko-scaffold holdout. We further add two controlled patent variants that explicitly suppress recurrent train pairs or seen-drug overlap. These analyses do not erase the benchmark-sensitivity finding. Instead, they sharpen it: under `patent_temporal_pair_novel` and `patent_temporal_drug_novel`, the AUPRC winner changes again, with `DTI-LM` overtaking `RF` while `RF - base` becomes unreliable on the pair-novel variant.

Table E6 Secondary real-OOD support benchmarks for the full five-model comparison panel. These panels use matched 3-seed evaluation, but remain supporting evidence because they reinforce rather than replace the temporal/provenance story established in the main text. Best mean AUPRC in each row is bold.

Benchmark	base (AUPRC/AUROC)	RF (AUPRC/AUROC)	DTI-LM (AUPRC/AUROC)	HyperPCM (AUPRC/AUROC)	GraphDTA-style (AUPRC/AUROC)
`BindingDB_Ki` / `unseen_target`	**0.6934 ± 0.0257** / **0.7192 ± 0.0068**	0.6328 ± 0.0465 / 0.6581 ± 0.0319	0.6648 ± 0.0177 / 0.6936 ± 0.0044	0.6515 ± 0.0301 / 0.6757 ± 0.0211	0.6280 ± 0.0237 / 0.6550 ± 0.0193
`DAVIS` / `unseen_target`	0.4097 ± 0.0275 / 0.8613 ± 0.0153	**0.4408 ± 0.0325** / **0.8806 ± 0.0128**	0.4090 ± 0.0570 / 0.8718 ± 0.0152	0.3840 ± 0.0269 / 0.8610 ± 0.0245	0.3601 ± 0.0502 / 0.8359 ± 0.0292

Table F7 Additional robustness analyses. These controls test provenance removal, stricter scaffold holdout, and tighter overlap restrictions within the patent benchmark. Values are mean ± std over three seeds.

Benchmark	Systems	Mean AUPRC / AUROC	Paired AUPRC CI	Read
BindingDB_nonpatent_Kd / nonpatent_temporal	base vs DTI-LM	base: 0.6369 ± 0.0173 / 0.7333 ± 0.0081; DTI-LM: **0.6531 ± 0.0066** / **0.7644 ± 0.0041**	+0.0163 [+0.0064, +0.0265]	temporal ranking difference survives without patent provenance
BindingDB_Kd / scaffold_drug	base vs HyperPCM	base: **0.6069 ± 0.0161** / **0.8351 ± 0.0037**; HyperPCM: 0.5983 ± 0.0368 / 0.8155 ± 0.0204	−0.0086 [−0.0203, +0.0025]	scaffold holdout removes the earlier HyperPCM edge
BindingDB_patent / patent_temporal_pair_novel	base, RF, DTI-LM	base: 0.7378 ± 0.0107 / 0.6813 ± 0.0115; RF: 0.7393 ± 0.0015 / **0.7114 ± 0.0024**; DTI-LM: **0.7491 ± 0.0076** / 0.7026 ± 0.0061	DTI-LM - RF: +0.0098 [+0.0061, +0.0134]	pair-novel control flips the AUPRC leader from RF to DTI-LM
BindingDB_patent / patent_temporal_drug_novel	base, RF, DTI-LM	base: 0.7255 ± 0.0117 / 0.6749 ± 0.0116; RF: 0.7356 ± 0.0016 / **0.7116 ± 0.0022**; DTI-LM: **0.7402 ± 0.0077** / 0.6989 ± 0.0069	DTI-LM - RF: +0.0047 [+0.0009, +0.0083]	drug-novel control still favors DTI-LM on AUPRC

Table G8 Paired bootstrap support for the five-model comparison panel. Positive deltas favor the second model in the comparison; negative deltas favor the first.

Comparison	Mean AUPRC delta	95% CI
`unseen_drug`: `RF - base`	+0.1830	[+0.1739, +0.1919]
`unseen_drug`: `RF - HyperPCM`	+0.1645	[+0.1558, +0.1733]
`blind_start`: `RF - base`	+0.0775	[+0.0546, +0.1030]
`scaffold_drug`: `HyperPCM - base`	−0.0086	[−0.0203, +0.0025]
`unseen_target`: `RF - base`	−0.0042	[−0.0175, +0.0084]
`patent`: `RF - base`	+0.0463	[+0.0441, +0.0484]
`patent`: `RF - DTI-LM`	+0.0409	[+0.0384, +0.0433]
`patent_pair_novel`: `RF - base`	+0.0015	[−0.0022, +0.0049]
`patent_pair_novel`: `DTI-LM - RF`	+0.0098	[+0.0061, +0.0134]
`patent_drug_novel`: `RF - base`	+0.0101	[+0.0065, +0.0137]
`patent_drug_novel`: `DTI-LM - RF`	+0.0047	[+0.0009, +0.0083]
`nonpatent_temporal`: `DTI-LM - base`	+0.0163	[+0.0064, +0.0265]
`nonpatent_temporal`: `RF - base`	+0.0832	[+0.0737, +0.0920]
`nonpatent_temporal`: `RF - DTI-LM`	+0.0670	[+0.0564, +0.0764]

Appendix G Paired uncertainty support

Table G8 summarizes the paired bootstrap intervals for the five-model comparison panel. The strongest supported deltas still include `RF` on synthetic `unseen_drug`, `blind_start`, and the aggregate patent and non-patent temporal benchmarks. At the same time, the controlled patent rows show why the paper should not overstate the temporal story. `RF - base` on `patent_pair_novel` crosses zero, while `DTI-LM - RF` is strictly positive on both controlled patent variants. By contrast, the synthetic `unseen_target` comparison `RF - base` also crosses zero, and the scaffold-split `HyperPCM - base` comparison crosses zero as well, which is exactly why those rows are described conservatively in the main text.

Table H9 Reusable benchmark artifacts released with each split. External prediction files can be scored without re-running the in-repo training code.

Artifact	Example files	Purpose
Split manifest	`manifest.json`	versioned metadata, split counts, prevalence, seed, and canonical paths
Partition tables	`train.csv`, `test.csv`	labeled partitions with `example_id` and exported overlap scores
Pair tables	`train_pairs.csv`	lightweight pair-only views for external pipelines
Entity tables	`drugs.csv` / `targets.csv`	unique drug and target vocabularies aligned to exported IDs
Full joined table	`all.csv`	complete resource snapshot for auditing or custom slicing
Standalone evaluator	`evaluate_prediction_csv.py`	scores a prediction CSV and returns global and overlap-bucket metrics

Appendix H Reusable benchmark resources

The benchmark release is intended to be reused independently of the in-repo models. Table H9 lists the concrete artifacts exported for each split and how they interact with the standalone evaluator. In practice, an external user only needs the canonical split bundle plus a prediction CSV keyed by `example_id`; the evaluator then reproduces the same global metrics and overlap-bucket metrics reported in this paper.

Appendix I Versioned release snapshot

The manuscript is tied to a fixed release snapshot so that reviewers can inspect the exact resource bundle used to generate the reported tables. The archived snapshot corresponds to git commit `a5d6b8b` and includes the exported split bundles under `benchmark_resources/`, the prediction and decision evaluators in `scripts/`, and the benchmark summaries under `reports/`. The companion manifest `benchmark_resources/RELEASE_MANIFEST.md` records the repository URL, full commit hash, and the concrete artifact paths needed to reproduce the paper tables from external prediction CSVs without rerunning the full training stack.

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